

Predicting Heart Failure Mortality

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Objective

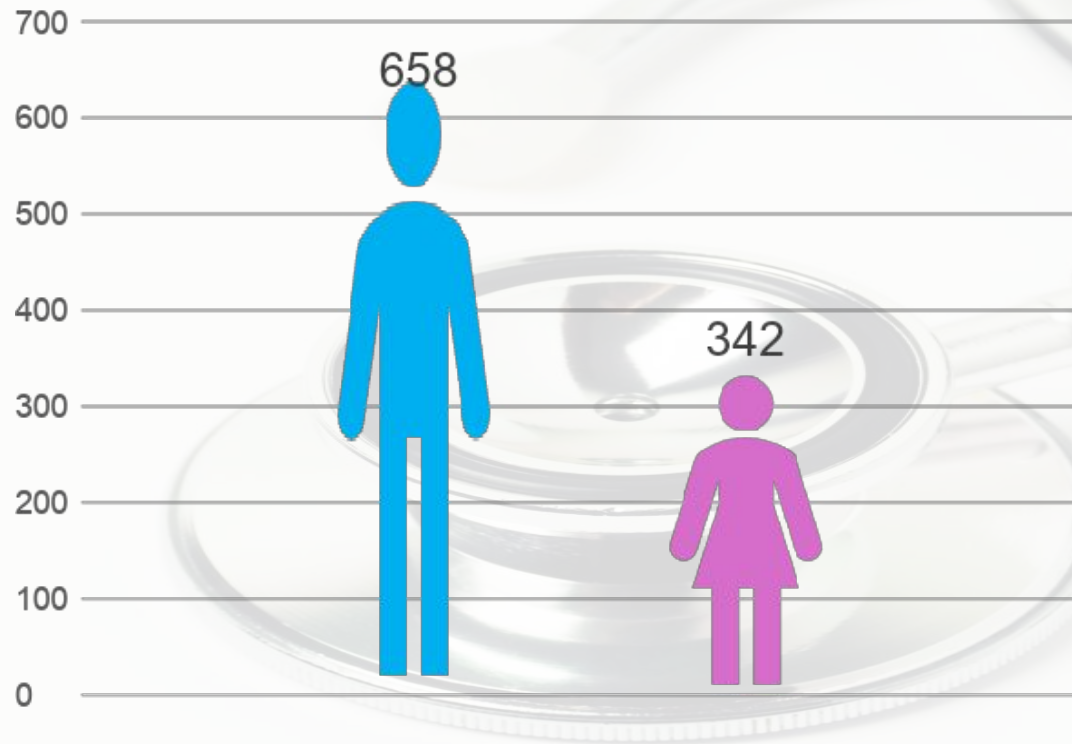


Achieve $\geq 75\%$ diagnostic accuracy with 95% confidence, outperforming Seattle Heart Failure Model's current 73.4% accuracy.



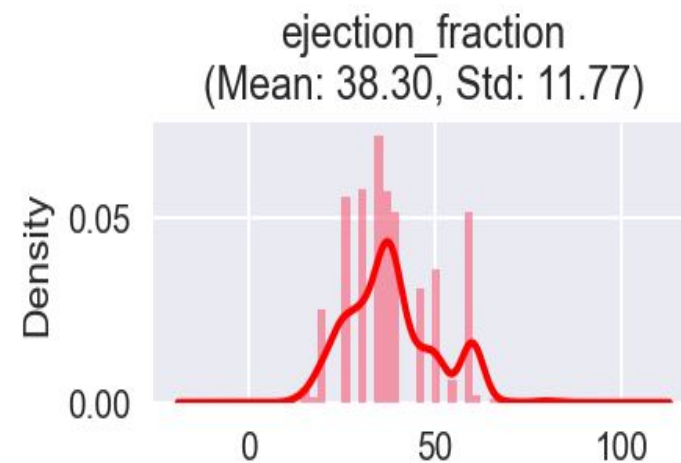
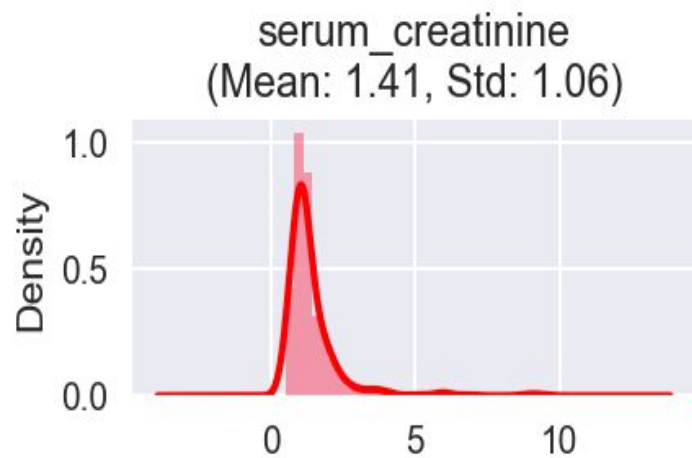
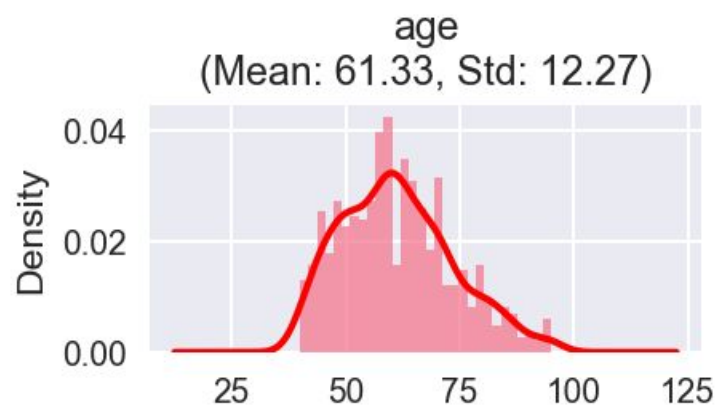
Even a 1% improvement is clinically meaningful because correctly diagnosing heart failure mortality risks earlier reduces hospital readmissions, decreases treatment delays, and saves lives.

UCI Dataset with Claude 4.5

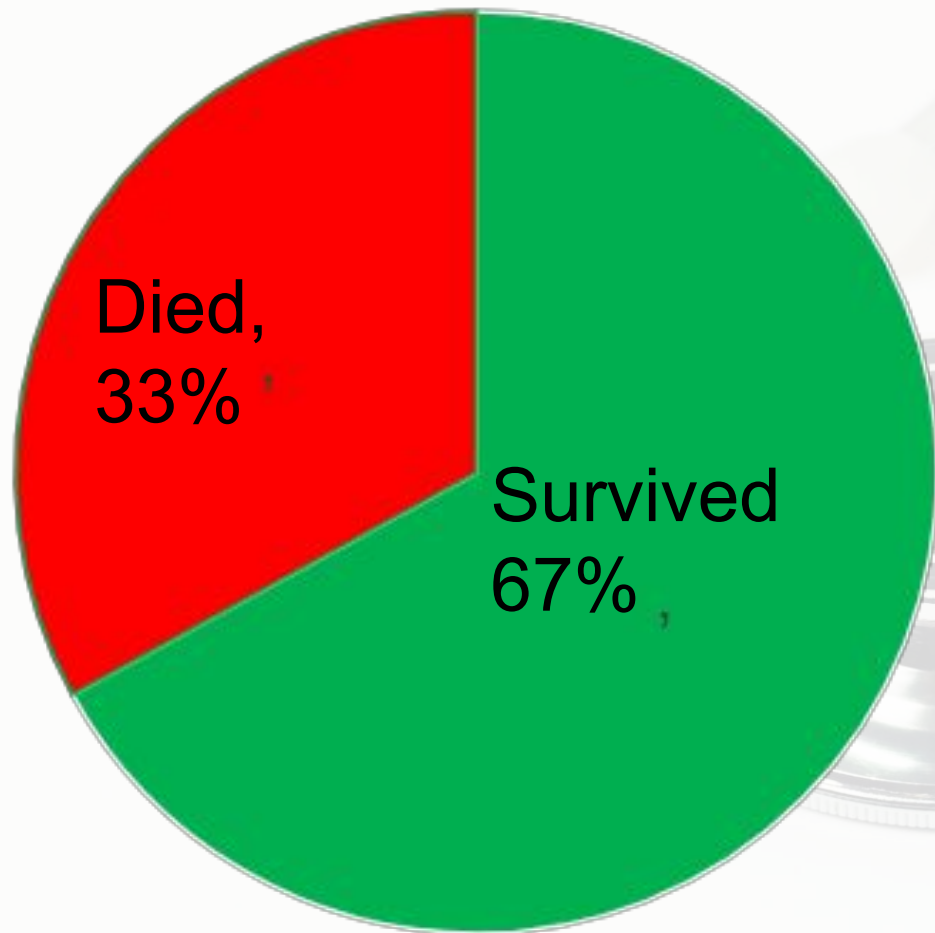


- age
- anemia
- creatinine phosphokinase (CPK)
- diabetes
- ejection fraction
- high blood pressure
- platelets
- sex
- serum creatinine
- serum sodium
- smoking
- time
- death event

Initial Exploration



Survival Distribution & Greatest Chance of Death



Anemia

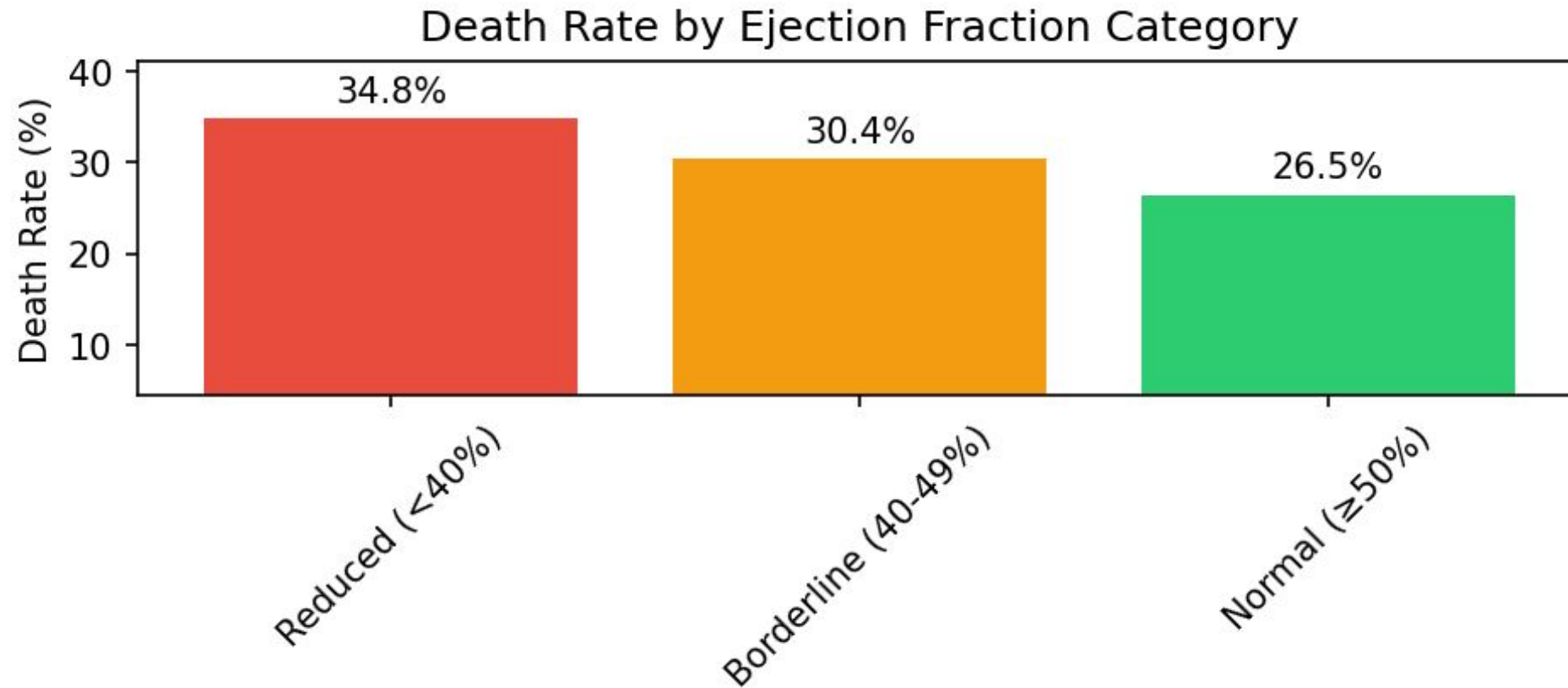


High Blood Pressure



Diabetes

Death Rate by Ejection Fraction



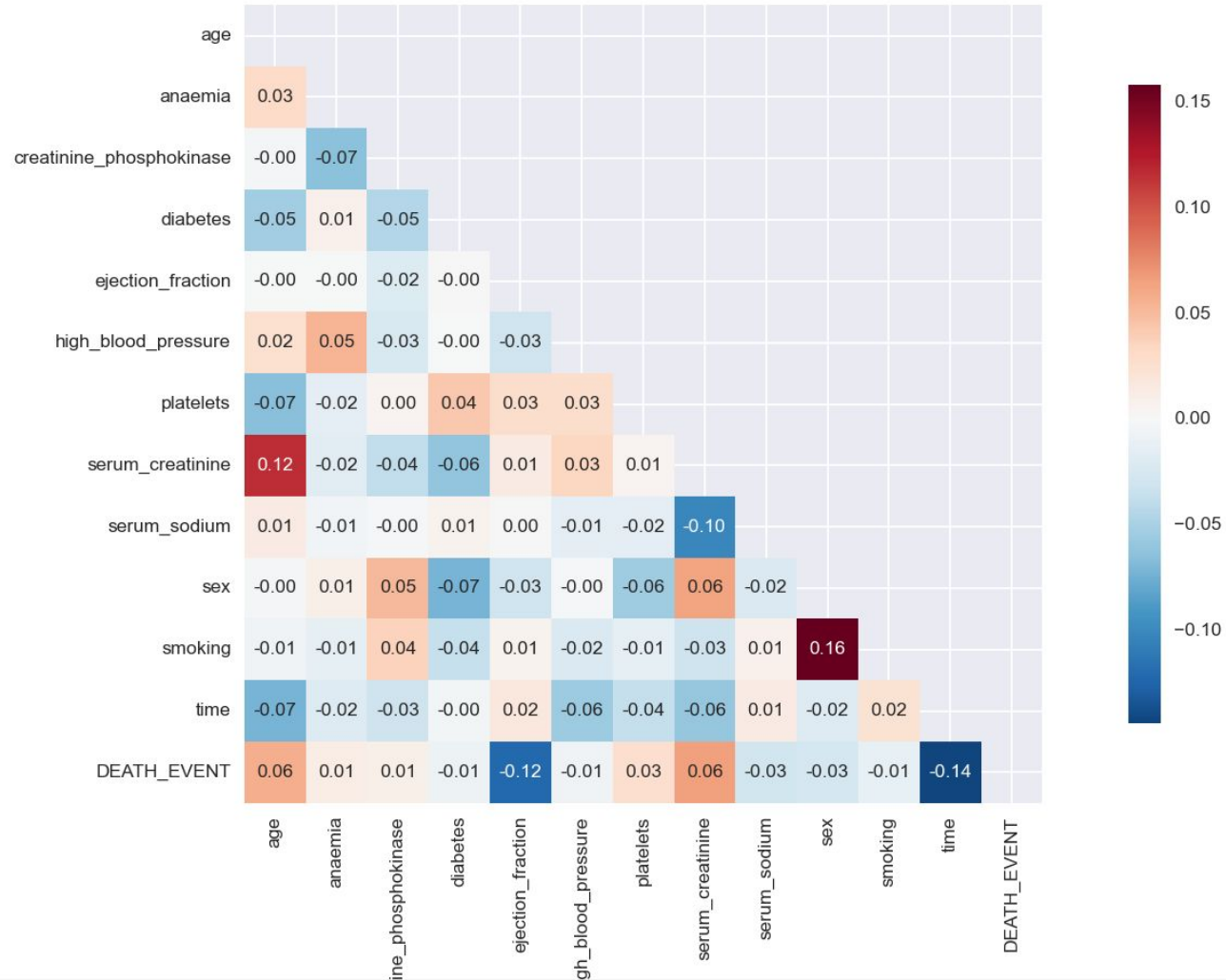
Most Correlative Features

There was three attributes with a p value less than 0.05:

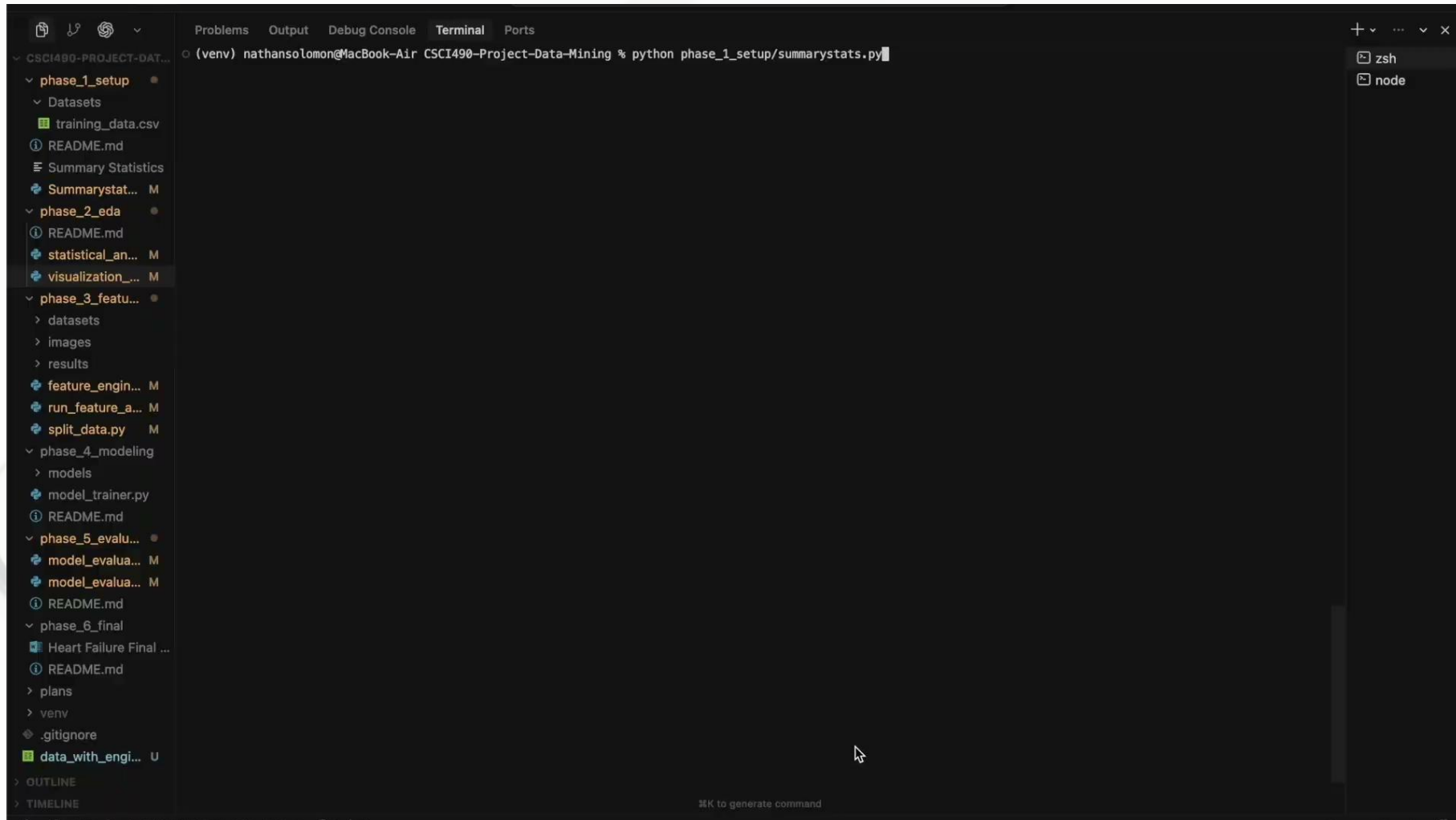
- time
- ejection fraction
- serum creatine

```
FEATURE CORRELATIONS WITH DEATH_EVENT:
age                r =  0.057 (p = 0.0725)
creatinine_phosphokinase r =  0.010 (p = 0.7605)
ejection_fraction  r = -0.122 (p = 0.0001) ***
platelets          r =  0.026 (p = 0.4138)
serum_creatinine    r =  0.065 (p = 0.0409) *
serum_sodium        r = -0.030 (p = 0.3472)
time               r = -0.145 (p = 0.0000) ***
anaemia            r =  0.012 (p = 0.7087)
diabetes           r = -0.008 (p = 0.7891)
high_blood_pressure r = -0.007 (p = 0.8269)
sex               r = -0.028 (p = 0.3846)
smoking           r = -0.012 (p = 0.6953)
```

Correlation Matrix for the Data



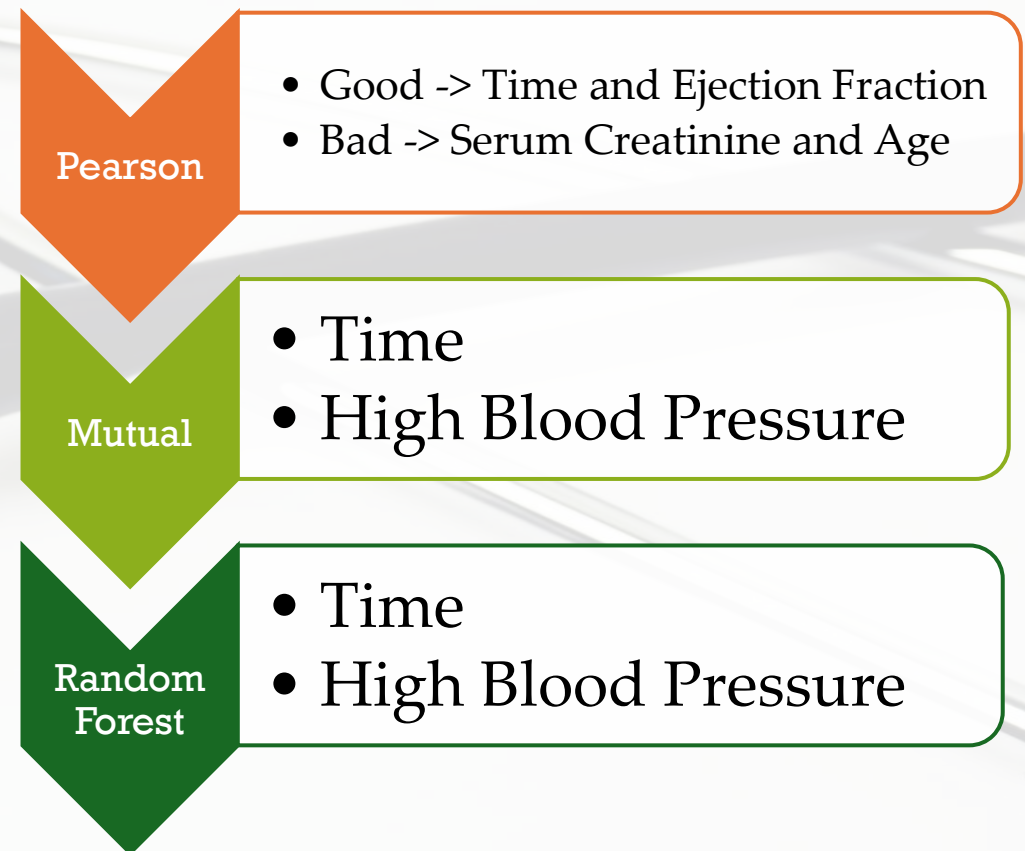
Demo Video



Feature Engineering & Selection

We applied three different feature selection methods to understand which variables are most important for predicting heart-failure mortality.

- Pearson correlation (linear relationships)
- Mutual Information (non-linear relationships)
- Random Forest importance (model-based)



These methods identified as important or not important

Pearson Correlation

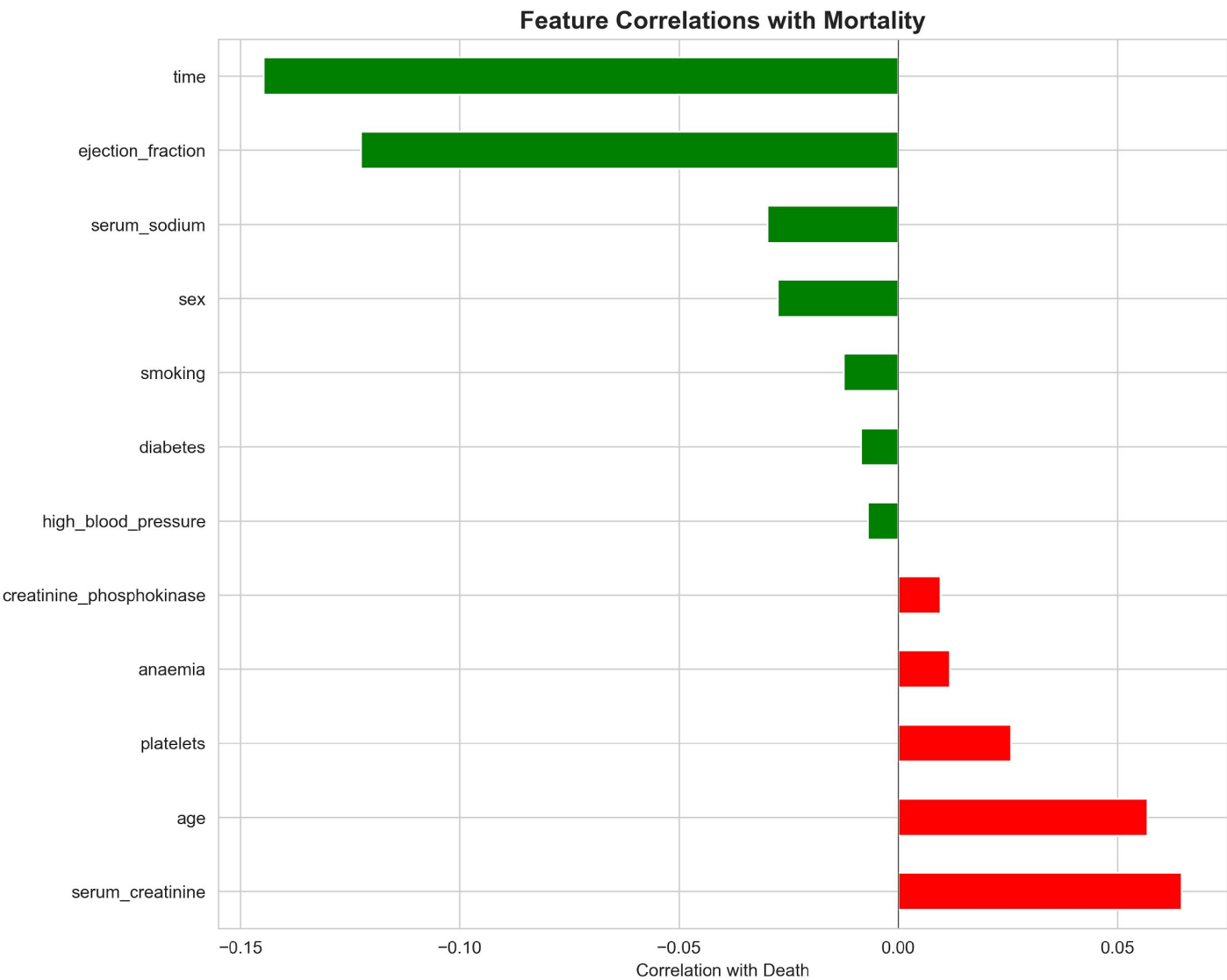
Pearson correlation measures linear relationships between each feature and death.

The two strongest protective factors (negative correlation with death) were:

- Time (longer follow-up tends to mean survival)
- Ejection Fraction

The strongest risk factors (positive correlation) were:

- Serum Creatinine
- Age



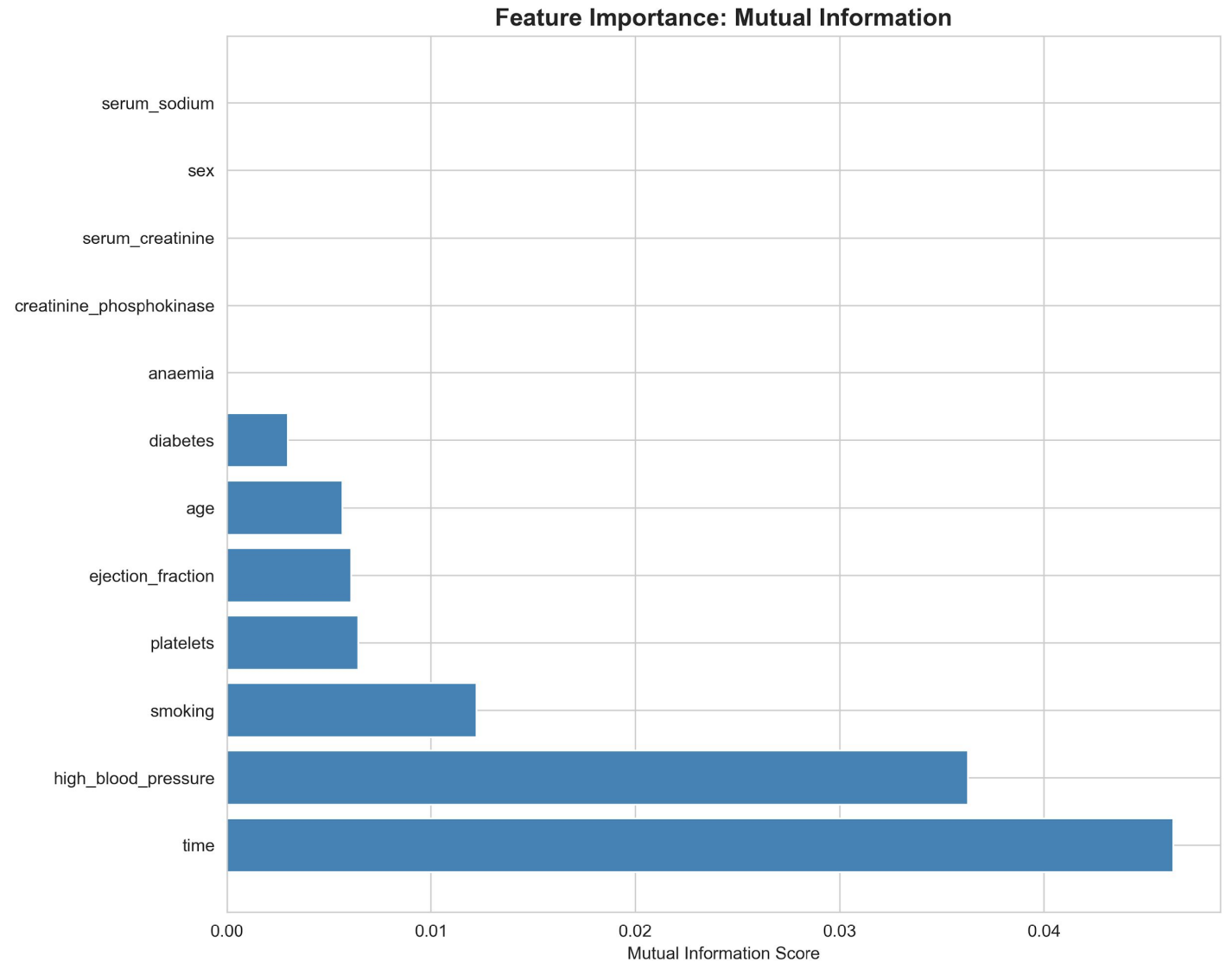
These results help identify which variables behave the most predictably in relation to mortality.

Mutual Score

Mutual Information captures both linear and non-linear relationships.

The highest MI scores were:

- Time
- High Blood Pressure
- Smoking (moderate)



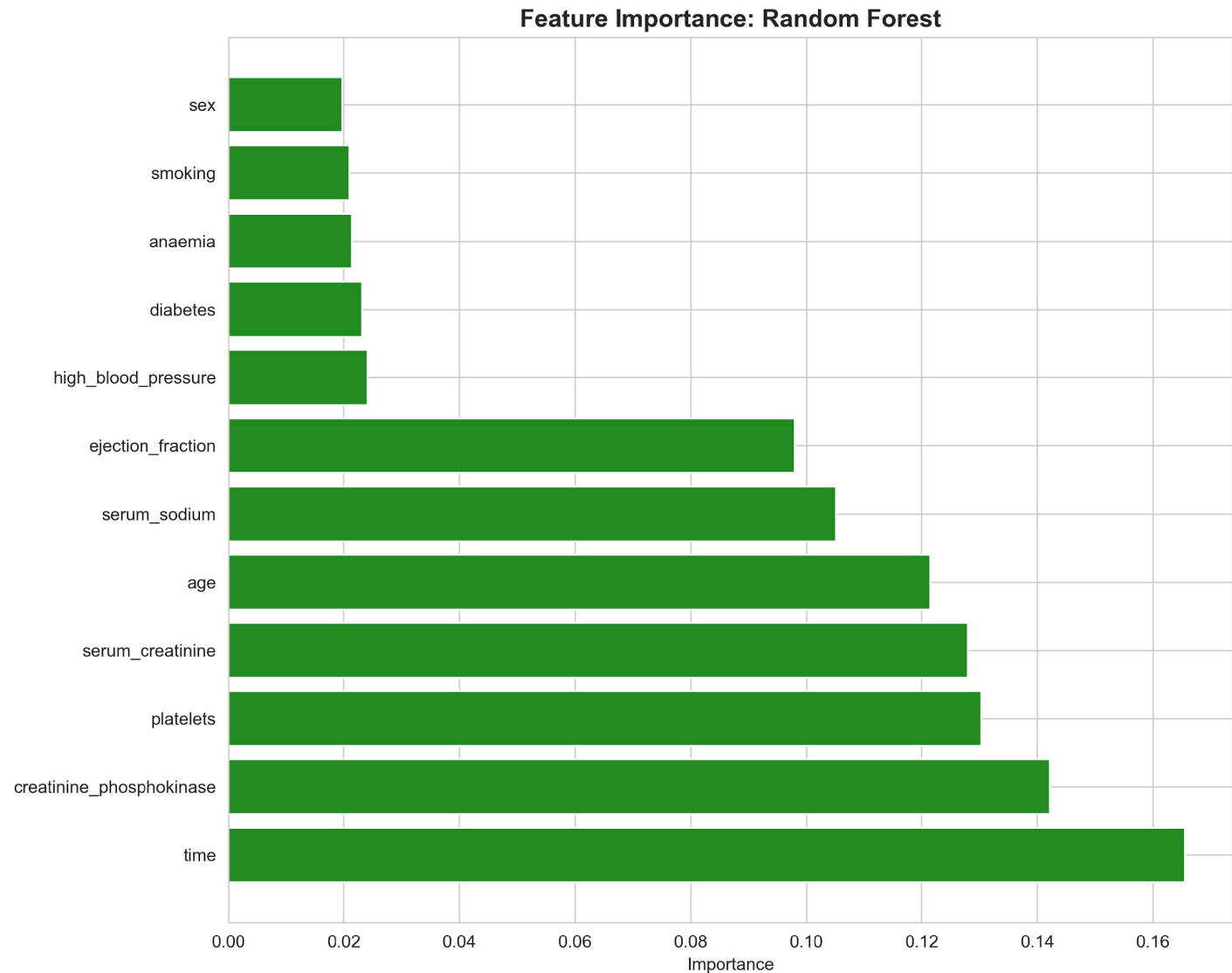
MI confirmed that “Time” is consistently informative, even outside linear effects.

Random Forest

Random Forest evaluates feature usefulness based on how often it improves decision splits.

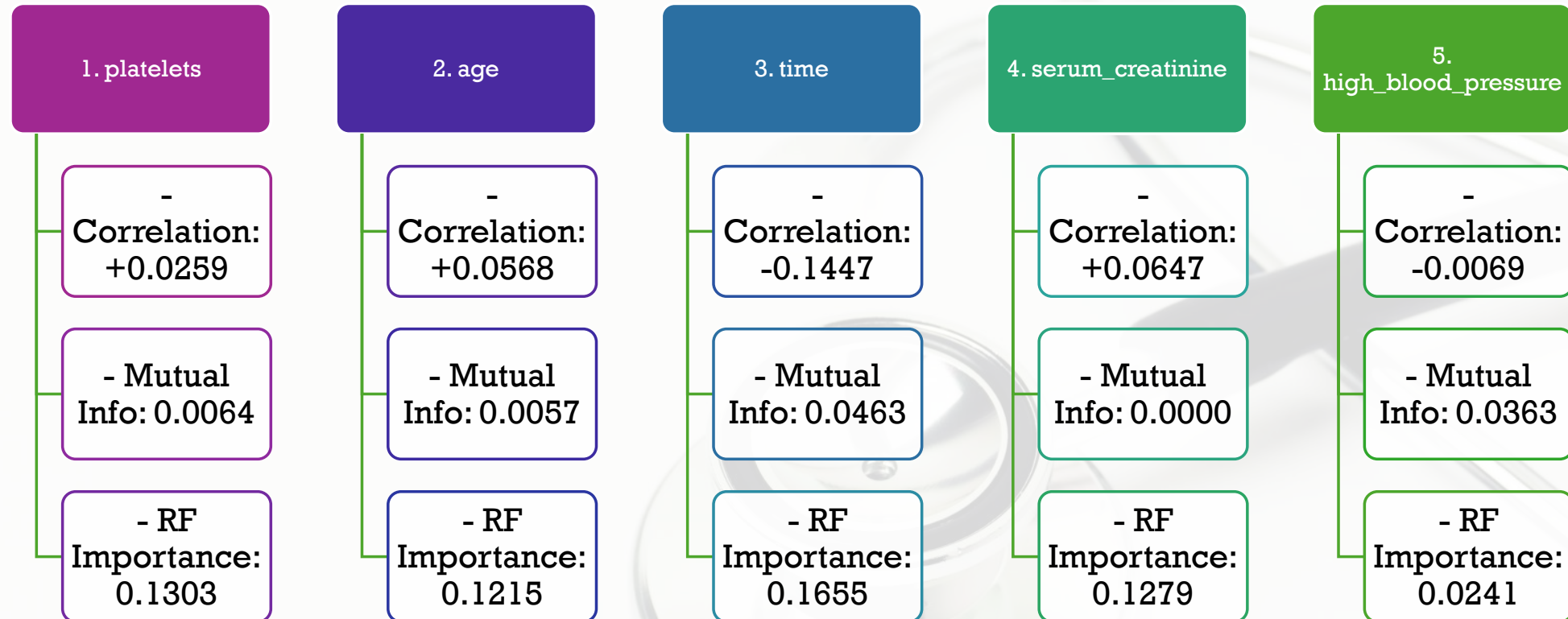
The most important features according to RF were:

- Time
- Creatinine Phosphokinase
- Platelets
- Serum Creatinine
- Age



This method provides a model-based perspective, highlighting features that improve predictive power.

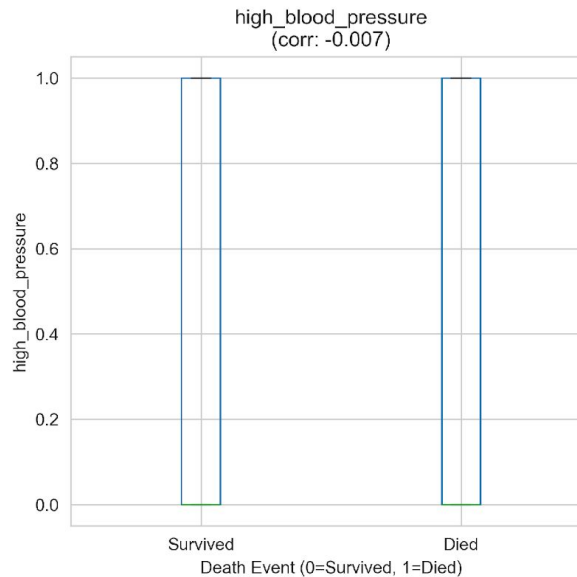
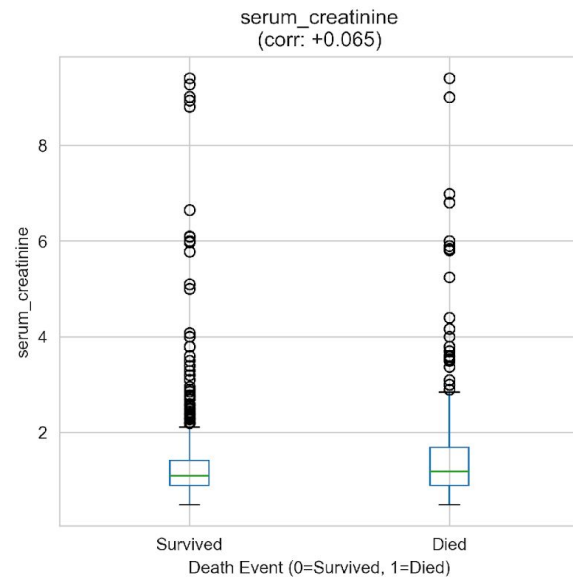
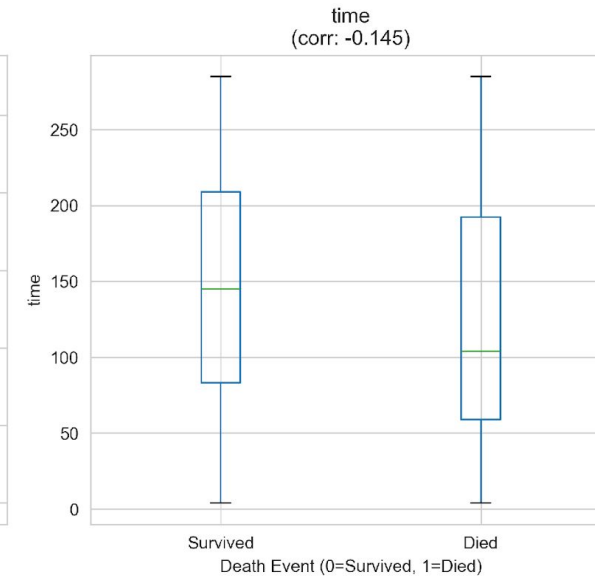
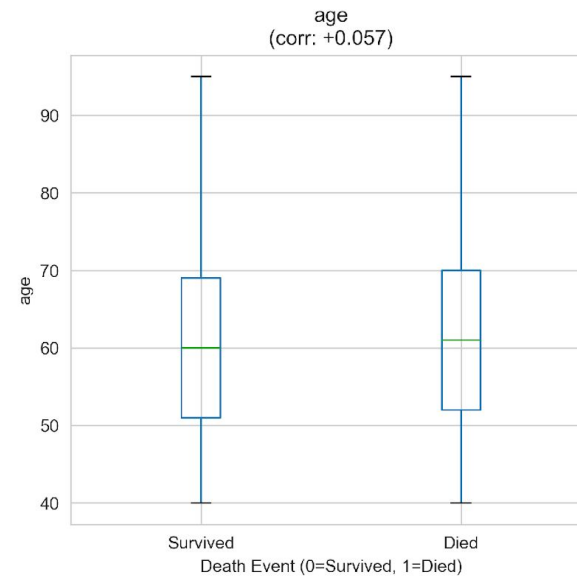
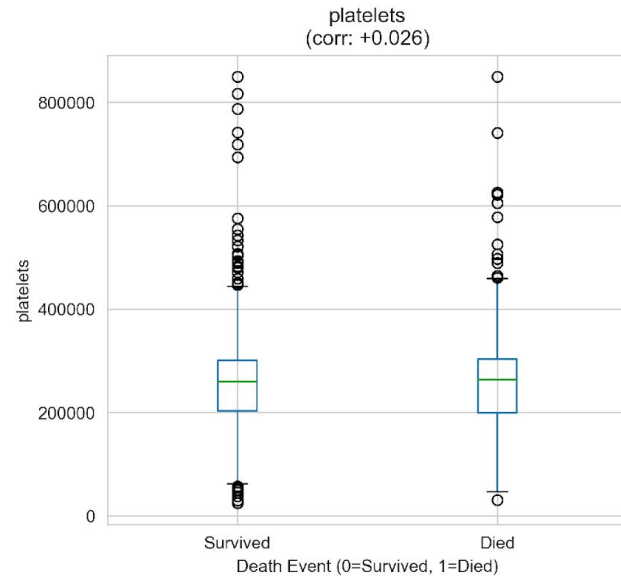
Top 5 Features



These features scored well across correlation, MI, and RF.
Time and Serum Creatinine also show strong clinical relevance.
These final features will be used in the modeling phase.

Top 5 Features Cont.

Top 5 Predictive Features by Mortality Outcome



Training models

Two datasets

- Original(13 features)
- Engineered(20 features)

Two algorithms

- Decision Tree(CART)
- Random Forest

- Use 80/20 split train/test
- Used StratifiedKfold to maintain 33% death rate across all folds
- Used GridSearchCV to perform 10-fold stratified cross validation for the parameter grids
- Conduct refitting to utilize all 800 samples from training set

Our Approach

Decision Tree(CART)



- Computationally inexpensive
- Fast classification
- Interpretable

Random Forest



- Robust
- More resilient to outliers and noise
- Handles high-dimensional data well

Process

- Total evaluations: $(576 \text{ DT} + 216 \text{ RF}) \times 2 \text{ datasets} \times 10 \text{ folds} = 15,840 \text{ model fits}$
- 4 fully-trained models

Results:

Cross-validation scores

Model	Mean CV Accuracy
Decision Tree(Original)	66.63%
Decision Tree(Engineered)	67.25%
Random Forest(Original)	68.25%
Random Forest(Engineered)	68.00%

Figure 1

- Decision Tree Search Space (576 combinations):
 - criterion: ['gini', 'entropy']
 - max_depth: [3, 5, 7, 10, 15, None]
 - min_samples_split: [2, 5, 10, 20]
 - min_samples_leaf: [1, 2, 5, 10]
 - max_features: ['sqrt', 'log2', None]
- Random Forest Search Space (216 combinations):
 - n_estimators: [50, 100, 200]
 - max_depth: [3, 5, 10, 15]
 - min_samples_split: [2, 5, 10]
 - min_samples_leaf: [1, 2, 5]
 - max_features: ['sqrt', 'log2']

Expectations



**AUC-ROC
scores greater
than 73.4**

Comparison of 4 models

OG Tree

• 0.608

EGN Tree

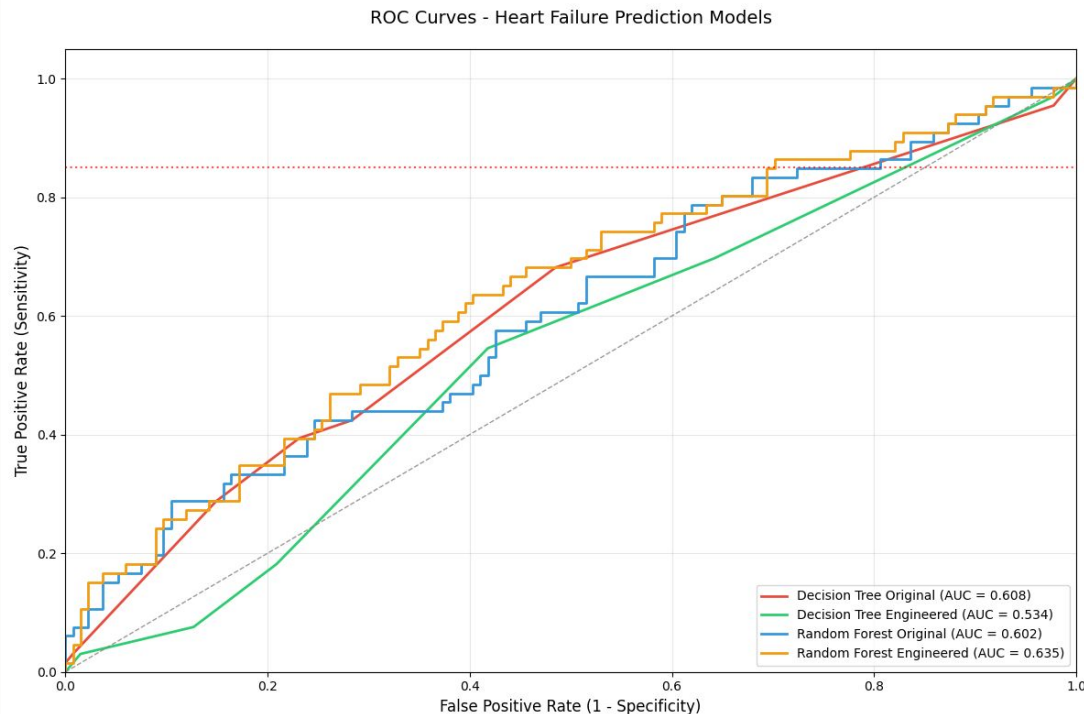
• 0.534

**OG Rand
FOREST**

• 0.602

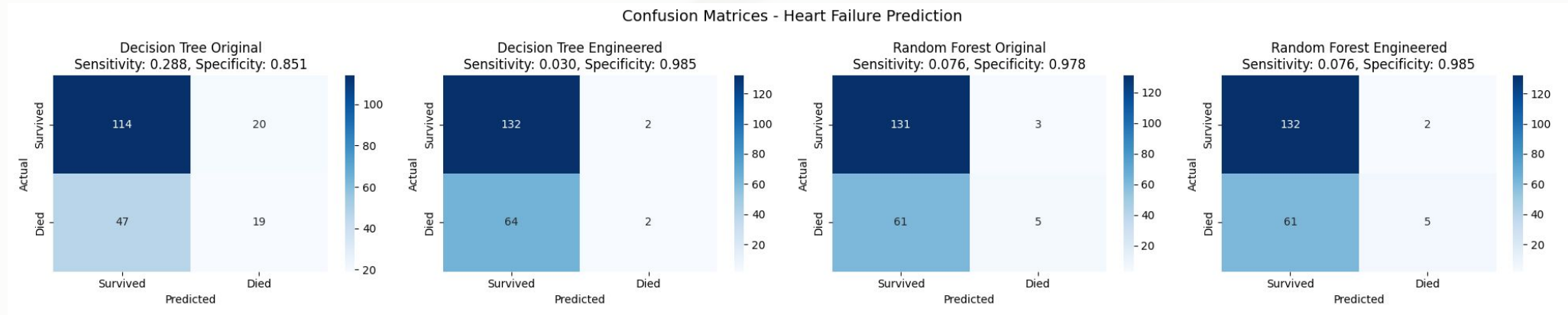
**EGN Rand
FOREST**

• 0.635



- The ROC graph on the left showcases 4 different models, Original and Engineered Decision Trees, as well as Original and Engineered Random Forest.
- Comparing all four models, engineering for decision trees create a worse result compared to Random Forest which engineering it boots the value.
- Despite our findings we found our values to be more in the middle, with the lowest being 0.534 and highest being 0.635. This essentially means the model's performance is no better than random chance at distinguishing between the two classes, suffering from a low performance.

Confusion Matrices with 4 models



- This confusion matrix showcases all 4 models and their values in this order: True Negatives, False Positives, False Negatives, and True Positives.
- The matrix showcases a 2:1 ratio, with about 67% of patients diagnosed with heart failure surviving.
- This matrix also shows a major problem, low sensitivity and high specificity.

The Good

- Great at predicting which patient will survive.

The Bad

- Poor performance to identify which patients will die

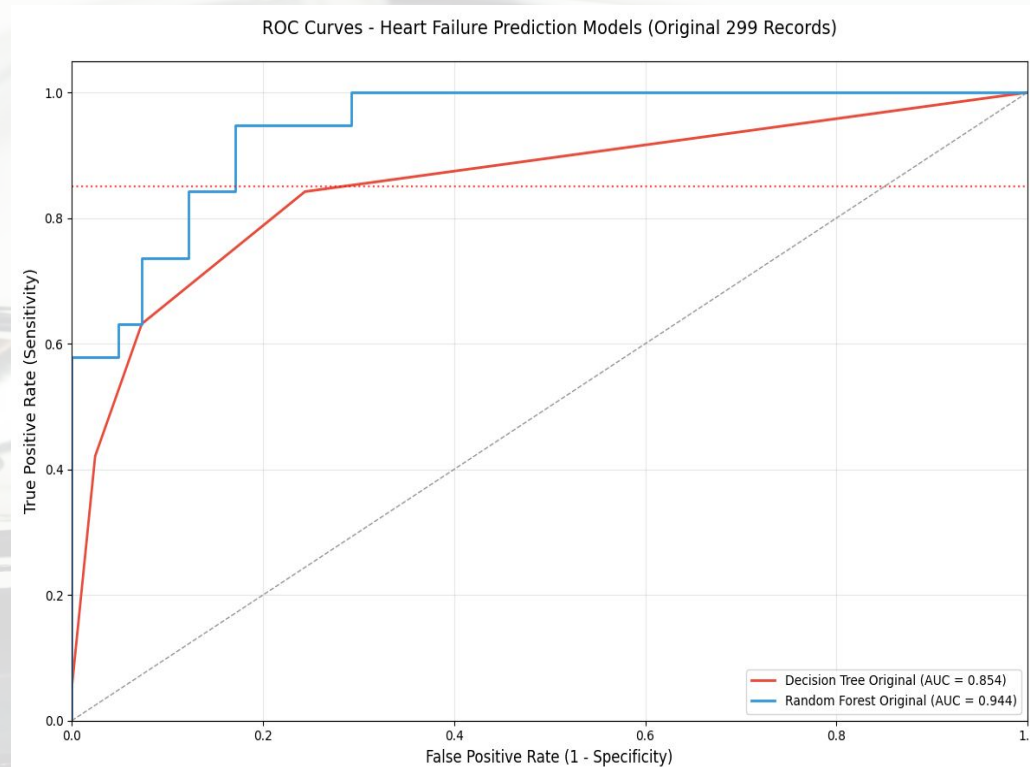
Model with Original Data

OG Tree

• 0.854

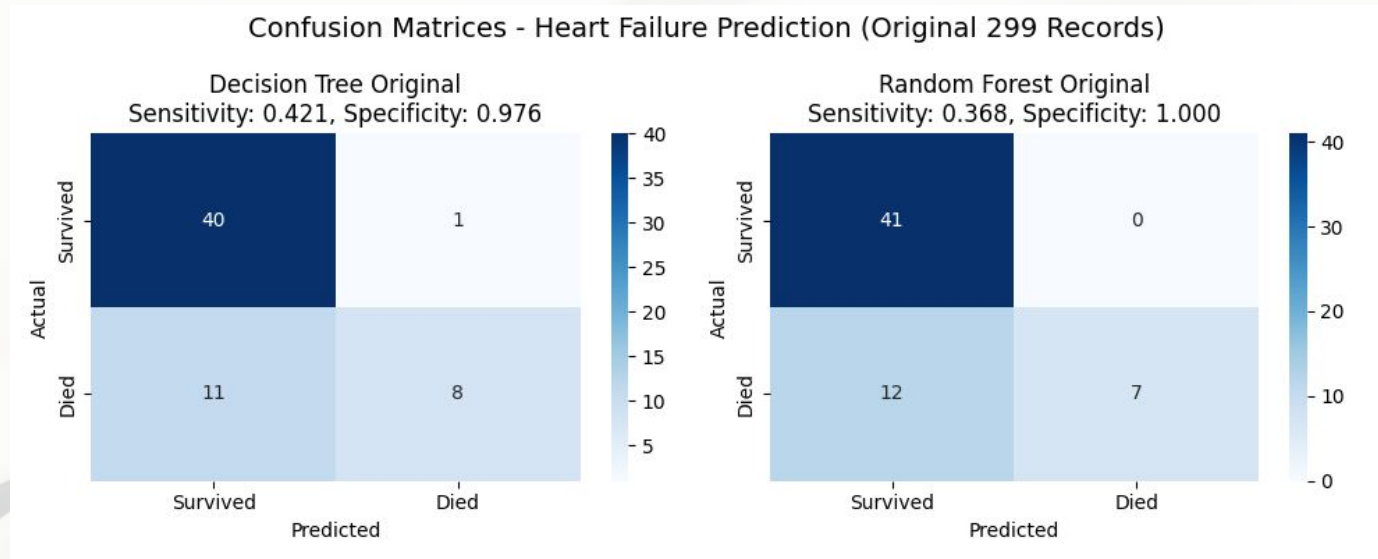
**OG Rand
FOREST**

• 0.944



- The ROC curves and AOC score in the “299” graph show how both models are close to 1.
- If we decide to compare this ROC graph with the full dataset one, we can see how the synthetic data skews the data due to the data being created “safely” as an average.

Confusion Matrices – Original Data



- This confusion matrix showcases the 2 models and their values in this order: True Negatives, False Positives, False Negatives, and True Positives.
- Despite being more accurate, as well as having one matrix say no patients have died, it still suffers the same problem of low sensitivity and high specificity, but it does have 80% accuracy based on patients being correctly classified, and has a balanced approach.

The Good

• Better accuracy and balanced

The Bad

• Still has issues identifying which patients will die

The Winning Model



73.4%

Seattle Heart Failure Model



68.5%

Our EGN Random
FOREST model



79.7%

Original 299 OG
Tree model

- Despite our best efforts, the Seattle Heart Failure Model ultimately out performs our model.
- Comparing our model with the original 299 values in the original data set, the original data clearly outperforms the hospital model by 6.3%.
- It's clear that using synthetic data seemed to have over-generated survival cases, as well as caused some overfitting within the decision trees and ultimately caused it to skew the values and results.

Conclusion and Future Work

Our goal: Accurately predict heart failure by 75% with a 95% confidence

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
<u>Setup</u> Started by increasing the dataset synthetically using Claude Initial Interest: age, serum_creatinine, ejection fraction based on research	<u>EDA</u> Basic statistics exploration such as mean and std deviation. 61.3 mean for age	<u>Feature Engineering</u> Pearson, Mutual Information Score, and Random Forest Classifier Created 7 new to increase size of 20 features	<u>Training Model</u> OG Decision Tree, OG Random Forest, Eng Decision Tree, Eng Random Forest Use KFold and hyper parameter tuning	<u>Evaluation</u> Use Bootstrap Use Confusion Matrices Low sensitivity and high specificity Eng Random Forest is best at 68.5%. With original dataset achieves 79.7%	<u>Future Work</u> Use different dataset with similar features instead of synthetic data.